

1. The question is about the interleaved execution of the two threads (Payroll and ATM withdrawal) presented in the lecture. However, in this case, the payroll threads starts first. If the initial balance B is \$200, what will be the final value of B after the interleaved execution of both threads.

B = \$200

• Interleaved Execution

- Load R_g, B
- add R_g, 1000

Interrupt

- Load R_g, B
- sub R_g, 80
- Store B, R_g

Return from int..

- Store B, R_g

1,200

2. True or False.

This question is related to the example using a payroll thread and an ATM withdrawal thread. If the two threads execute concurrently in an interleaved way (i.e., instructions from each would execute alternatively), the integrity of the shared variable B (balance) may get violated.

True

3. Mutual exclusion means _____.

A thread will not manipulate some shared data while another is also manipulating the same data D.

4. Based on the textbook (Section 6.2), when a thread is not in its critical region (section), it is in its _____ section.

remainder

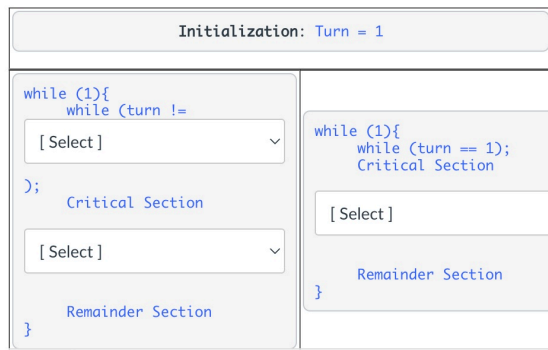
5. A solution to mutual exclusion must have the following components:

Entry code
Initialization
Exit Code

6. A solution to mutual exclusion must have the following components: initialization, entry code, and _____ code.

exit

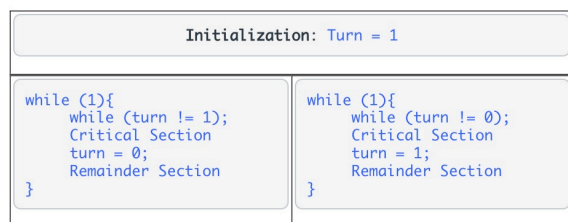
7. Select the best option to complete code for the first attempt to provide mutual exclusion. Let us limit variables to take only value 0 or 1.



1
turn = 0
turn = 1

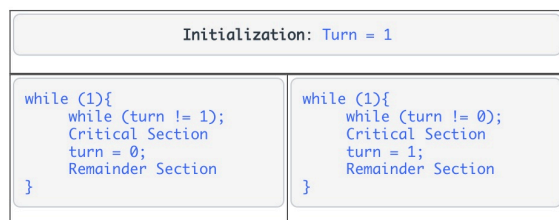
8. A busy waiting solution means that _____.
the entry code uses the CPU to wait before access to the critical region

9. The solution below does NOT fulfill the progress requirement because:



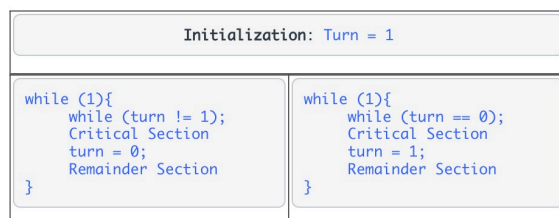
If one of the threads dwells indefinitely in the remainder section, it will eventually block the other thread from ever entering its own critical region

10. The solution to mutual exclusion below meets these requirements:



Mutual exclusion

11. The solution providing mutual exclusion below meets these requirements:



None of these answers